

## Introduction

**Surprise minimization** was first described in the field of neuroscience and has since been incorporated into **active inference models**, in which individual decision-making contributes to collective behavioral patterns. Surprise minimization is the process by which individuals act to reduce the difference between the sensory information they process and their expectations of the state of their environment. In the model, this refers to the observed and expected distances between each individual and their neighbors. In nature, uncertainty is common and can interfere with these processes; however, incorporating uncertainty into models via **noise** can more accurately represent emerging behavioral patterns under **different sensory conditions**. A better understanding of how individuals behave and respond to these sensory conditions can support population predictions amid environmental change, accelerate hypothesis testing, and guide field research experiments.

## Model Exploration

This project extended an existing active inference model to explore:

- Changes in model robustness and sensitivity
- Generalizing the model to represent noisier environments
- The addition of stochastic interactions among agents

## The Model

The core model of this project has been adapted from the work of Conor Heins et al., which previously demonstrated that collective behavioral patterns can arise in models driven by surprise minimization. This model can be separated into the generative process and the generative model, in which a focal agent's observations and predicted observations from its belief are compared, and the resulting prediction error is reduced by minimizing free energy (*Figure 1*). Environmental noise is also incorporated into the model, impacting both the focal agent's sensory input [ $Z_H$ ,  $Z_{H'}$ ] and realized movement [ $Z_{Action}$ ].

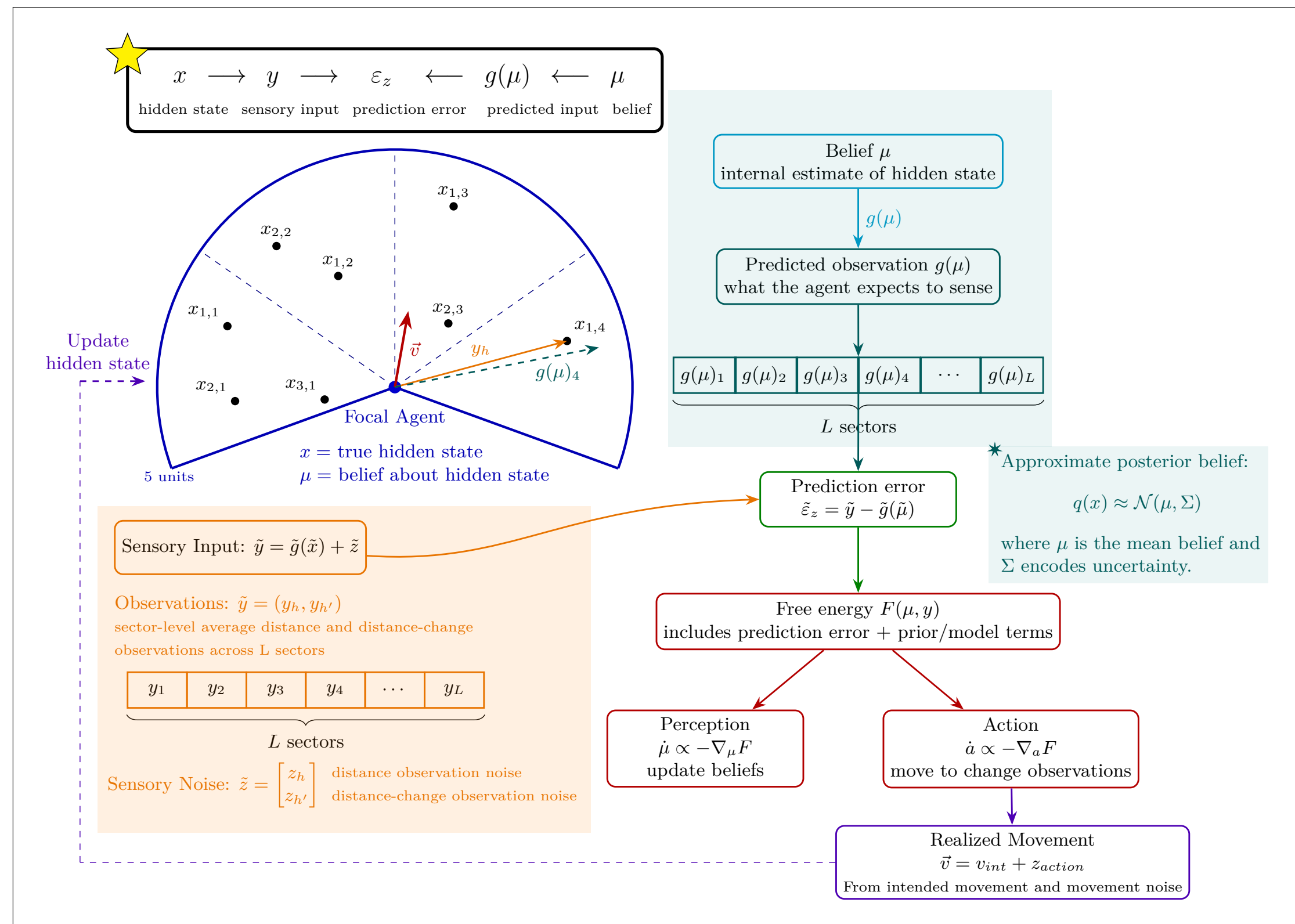


Figure 1. A visualization of the active inference model for collective behavior. The left side (orange box) describes the generative process – how the simulated world generates the focal agent's sensory observations – and the right side (teal box) describes the generative model – the focal agent's internal beliefs.

The processes in this model contribute to different collective behavioral patterns which are explored in this project and categorized as:

- Polarized** – coherent motion with a unified direction and strong velocity correlations
- Milling** – circling motion with a high angular momentum around the group center of mass
- Disordered** – no strong organizational patterns (neither polarized nor milling)

These behavioral patterns are determined via a comparison between the polarization and angular momentum values as described in *Table 1*.

| Behavioral Pattern | Polarization | Angular Momentum |
|--------------------|--------------|------------------|
| Polarized          | High         | Low              |
| Milling            | Low          | High             |
| Disordered         | Mid-to-Low   | Mid-to-Low       |

Table 1. The collective behavior patterns explored by the model (Originally described in Heins et al. (2024)).

## Quantifying Collective Behavior

The **group polarization** [ $\hat{p}$ ] is calculated via a time average of the group velocity alignment [ $p(t)$ ] as described in *Equation 1*. As the speed is fixed in this model, the velocity [ $v_i(t)$ ] does not need to be normalized in *Equation 2*.

$$\hat{p} = \frac{1}{T-t_0} \sum_{t=t_0}^T p(t) \quad (1)$$

$$p(t) = \frac{1}{N} \left\| \sum_{i=1}^N v_i(t) \right\| \quad (2)$$

The **group angular momentum** [ $\hat{m}$ ] is calculated via temporal and group averaging of the individual angular momentum contributions [ $m(t)$ ] around the group center of mass [ $c$ ] as described in *Equation 3*. The relative position vector for each agent [ $r_{ic}(t)$ ] points from the center of mass to that agent's position  $\left[ \frac{r_i - c}{\|r_i - c\|} \right]$ .

$$\hat{m} = \frac{1}{T-t_0} \sum_{t=t_0}^T m(t) \quad (3)$$

$$m(t) = \frac{1}{N} \left\| \sum_{i=1}^N r_{ic}(t) \times v_i(t) \right\| \quad (4)$$

## Changes to the Model & Parameters

### Separating the Noise Realization from Initialization

The original model randomized the noise and the initial conditions using the same seed parameter. To explore the model's robustness under different noise values, a new branch was created with separate seed parameters for the initialization and noise realization.

### One Random Neighbor per Sector

The original model averaged the neighbor distances in each occupied sensory sector. To add stochastic interactions, a new branch was created where the distance to a random neighbor in each occupied sensory sector was selected at each time step as the observation for that sector. This random neighbor selection is encoded by the noise-realization seed.

### Parameter Sweeps

Much of the model experimentation was explored via changing the noise values (*Table 2*) to consider noisier environments and explore the model's sensitivity to different noise types.

| Parameter    | Description                                      |
|--------------|--------------------------------------------------|
| $Z_H$        | Observation noise on neighbor distance           |
| $Z_{H'}$     | Observation noise on change in neighbor distance |
| $Z_{Action}$ | Movement noise                                   |

Table 2. The generative process noise parameters

## Comparing Noise Sensitivity

The group polarization and angular momentum values were recorded over 10-second runs at different  $Z$ -noise value from 0.001 to 0.5, as shown in *Figure 2*. Upon comparison, the model appears to be the most sensitive to changes in  $Z_{Action}$  [ $Z = 0.001, 0.01$ , and  $0.5$ ; L2 distances: 0.14775, 0.14879, 0.10001] and the most robust under changes to  $Z_H$  [ $Z = 0.001, 0.01$ , and  $0.5$ ; L2 distances: 0.00001, 0.00002, 0.00005]. The presence of  $Z_{H'}$  and  $Z_{Action}$  noise resulted in greater fluctuations in angular momentum over time compared to the presence of  $Z_H$  noise.

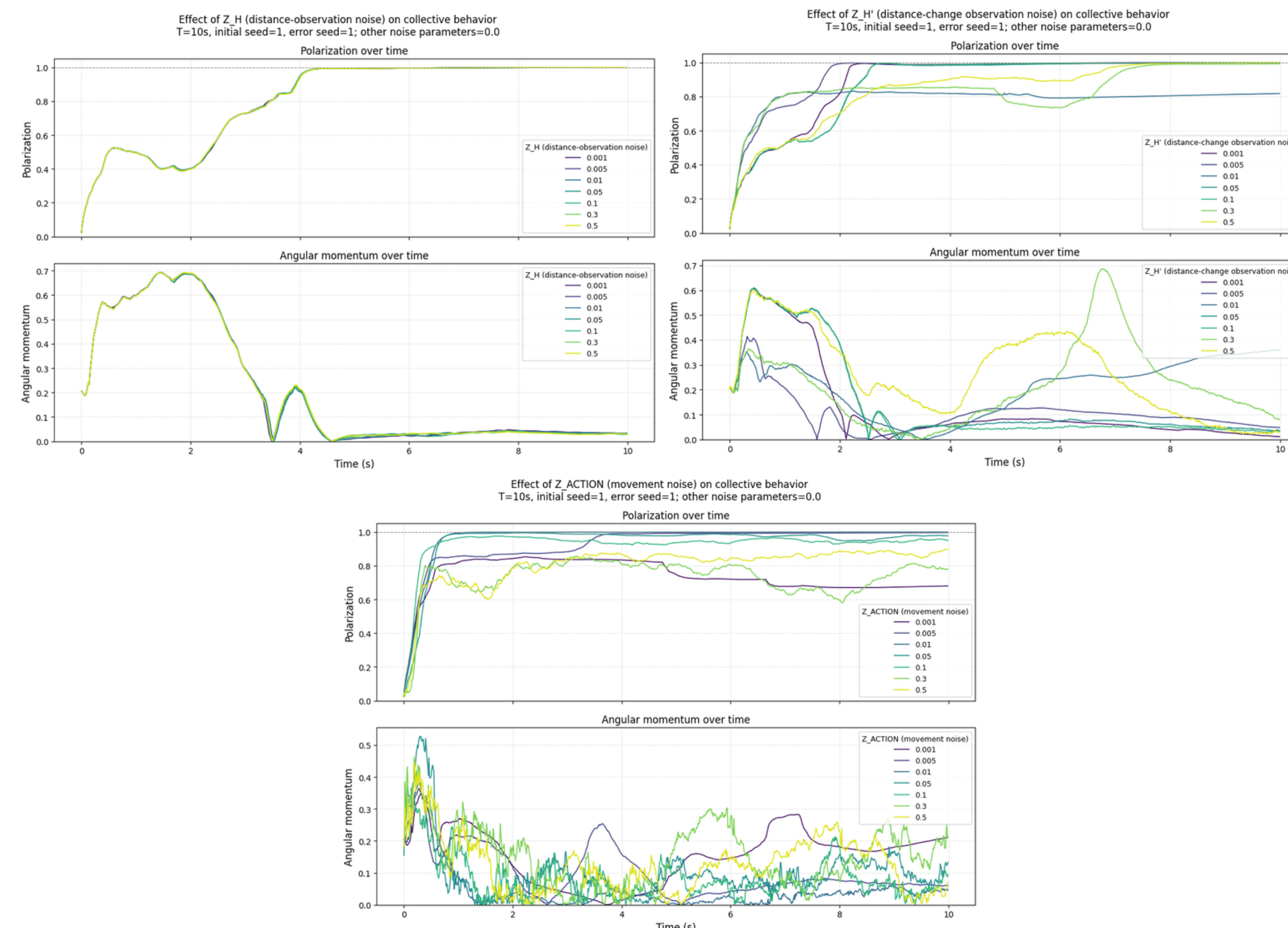


Figure 2. The group polarization and angular momentum across different  $Z$  noise parameter values [clockwise from top left:  $Z_H$ ,  $Z_{H'}$ , and  $Z_{Action}$ ]. Plotted runs lasted 10 seconds and only one noise parameter type was active per run.

## Stochastic Neighbor Interactions

The average polarization and angular momentum values were compared across ten noise realizations with different noise values to explore the impact of stochastic decision making on group behavioral outcomes (*Figure 3*). Specifically, the baseline model was compared with the "One Random Neighbor per Sector" branch update to observe changes in the model robustness and determine whether divergence would occur with the presence of stochastic interactions. In *Figure 3*, the standard deviation regions overlap across all  $Z$ -noise values (from 0.001 to 0.5) between the two model versions, indicating the model is robust under the presence of some forms of stochastic neighbor interaction. Notably, the presence of this stochasticity resulted in a much lower and more consistent angular momentum across  $Z_{H'}$  values compared to the baseline model.

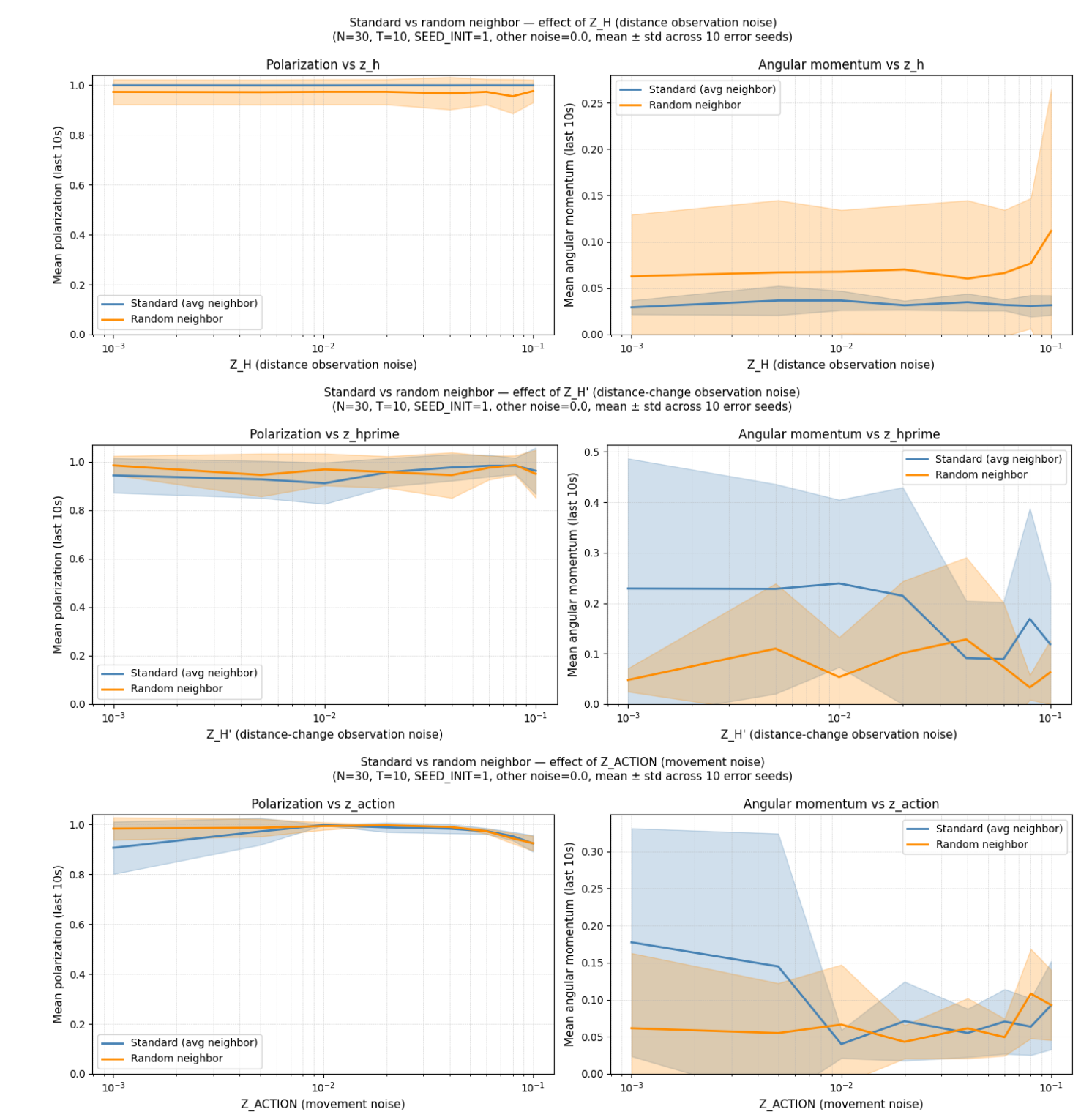


Figure 3. The average polarization and angular momentum values across ten noise realization seeds at different  $Z$  values for the average and random neighbor branches (from the top down:  $Z_H$ ,  $Z_{H'}$ , and  $Z_{Action}$ ). The shaded portions of the graphs represent a  $\pm 1$  standard deviation region.

## Discussion

### Key Findings:

- Active inference models respond differently depending on what stage of the process uncertainty enters (*Figure 2*).
  - The model acted more **robust** under changes in  $Z_{H'}$ , which occurs **before inference**.
  - The model was **more sensitive** to changes in  $Z_{Action}$  which is incorporated **after inference**.
  - Angular momentum fluctuations across  $Z_H$  values (*Figure 3*) suggest the ability to **accurately observe the distance-change rate of neighbors impacts rotational stability** more than accurately observing distance.
- Stochastic interactions** between neighbors does **not appear to interrupt the polarization** tendencies of the model (*Figure 3*).

### Future Work:

- Comparison between average and random neighbor plots across different sensory distances
  - Does divergence occur across visibility conditions?
- Separating the random neighbor assignment and noise realization parameters
- Performing a numerical analysis on discussed results to statistically confirm differences beyond characteristic observations
- Further iterations across different initializations, noise realizations, and noise values.

## Conclusion

- Fish-like agents can maintain organized collective behavior despite imperfect sensory observations
  - Further supports active inference as a possible framework for studying fish schooling
- Motivate hypothesis that physical perturbations (i.e. turbulence) result in higher rates of disrupted schooling patterns in fish-like agents.
- Expanding this model to accurately display fish responses to sensory conditions can support:
  - Predictions of schooling behavioral differences under environmental changes
  - Guiding field research directions and testing hypotheses

## Supplemental Information

### Acknowledgments

This project was supported by the Cal-Bridge Summer program and the University of California, Riverside Mentoring Summer Research Internship Program (MSRIP). I am deeply grateful for this support. I would also like to thank my advisor Dr. Siting Liu for her guidance and support during this project.

